

From Theory to Desk: Quant Tricks That Actually Matter

A Practical Guide to Greeks, Vol Surface/Smiles, YC in IR, Time-Series Models, Numerical Methods & Monte Carlo and Risk/XVA

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Where f is transition density. **Pros:** Works for any payoff (including discontinuous). **Cons:** Higher variance, needs density known.

Greek Method Selection "Smooth payoff $\rightarrow$ Pathwise. Discontinuous $\rightarrow$ Likelihood. Lazy $\rightarrow$ Bump" Pathwise: $\partial V / \partial \theta = \mathbb{E}[\partial \text{Payoff} / \partial \theta]$ "I use pathwise for Deltas of vanillas. Bump for Gammas of barriers (second derivative). Likelihood for digital Greeks."

Part 3: Trees & PDE Tricks

7. Binomial/Trinomial Trees: The OG Lattices

Risk-Neutral Pricing:

$$C = e^{-r\Delta t}(pC_u + (1-p)C_d)$$

Cox-Ross-Rubinstein (CRR) Parameters:

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = \frac{1}{u}, \quad p = \frac{e^{r\Delta t} - d}{u - d}$$

"Tree Probabilities Must be Probabilities" **Memory:** $0 < p < 1$ is **no-arbitrage condition**. If p or $1 - p$ negative $\rightarrow$ your step size is too big. **Fix:** Reduce Δt or switch to trinomial.

Trinomial Tree (Hull-White):

Up, Middle, Down with probabilities p_u, p_m, p_d

Advantage: Better stability, handles mean-reverting processes.

Stability Condition:

$$\Delta t < \frac{\Delta x^2}{\sigma^2} \quad (\text{for explicit PDE/trees})$$

Tree Checklist "No negative probs, stability condition, correct discounting" $p = (e^{r\Delta t} - d)/(u - d)$ "I use trees for American options. Check $p \in (0, 1)$ at every node. If not, increase steps."

8. Finite Difference PDE: The Grid Master

Black-Scholes PDE:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

Discretization:

$$\frac{V_i^{n+1} - V_i^n}{\Delta t} + \frac{1}{2}\sigma^2 S_i^2 \frac{V_{i+1}^n - 2V_i^n + V_{i-1}^n}{\Delta S^2} + \dots = 0$$

Explicit Scheme:

$$V_i^{n+1} = aV_{i-1}^n + bV_i^n + cV_{i+1}^n$$

Pros: Simple. **Cons:** Conditionally stable (Δt tiny).

Implicit Scheme:

$$aV_{i-1}^{n+1} + bV_i^{n+1} + cV_{i+1}^{n+1} = V_i^n$$

Pros: Unconditionally stable. **Cons:** Solve linear system.

Crank-Nicolson:

$$\frac{V^{n+1} - V^n}{\Delta t} = \frac{1}{2} \left(\text{RHS}^n + \text{RHS}^{n+1} \right)$$

Memory: Average of explicit and implicit. **Second-order accurate**, stable.

"Explicit is Like a Tightrope, Implicit is Like a Floor"

digitals) but higher variance. **Rule:** Pathwise for continuous, LR for discontinuous. Bump-and-reprice for second derivatives (Gamma) where both fail."

6. When do you use Quasi-Monte Carlo? Pricing a 5-asset basket. **Answer:** "QMC (Sobol, Halton) gives $O(1/N)$ convergence vs $O(1/\sqrt{N})$ for MC. Use when:

- **Dimension moderate:** 2-10 factors. QMC beats MC here.
- **Smooth integrand:** Payoff is continuous (not digital).
- **High accuracy needed:** Eg, CVA pricing where error must be $< 0.1\%$.

Caveat: QMC doesn't allow incremental path addition (no error estimate). I run both MC and QMC: MC for daily risk (fast), QMC for final pricing (accurate)."

7. How do you price an American option with MC? Longstaff-Schwartz method. **Answer:** "Least-Squares Monte Carlo (LSM):

1. Simulate N paths to expiry.
2. At each exercise date t , identify in-the-money paths.
3. **Regression:** Regress discounted continuation value on state variables (e.g., S_t, S_t^2) for ITM paths.
4. **Exercise rule:** Exercise if payoff $>$ regressed continuation value.
5. **Value:** Discount final cash flows back.

Keys: Use robust regressors (Laguerre polynomials). Run 2-pass: first pass estimates exercise strategy, second pass prices (to avoid bias)."

8. What's wrong with using uniform Δt for barriers? MC misses barrier hits between steps. **Answer:** "Discretization bias. If barrier is at 80 and you step daily, spot could jump from 81 to 79 **between** steps, but MC misses it. Fix: Brownian bridge technique. For each step, compute probability of hitting barrier conditional on start and end:

$$P_{\text{hit}} = \exp\left(-\frac{2(S_t - B)(S_{t+1} - B)}{\sigma^2 S_t^2 \Delta t}\right)$$

Draw uniform and compare. This removes first-order bias. Alternatively, use PDE for barriers."

9. How do you handle high-dimensional MC? Basket of 50 stocks. **Answer:** "Three tricks:

1. **Factor reduction:** Model 3 principal components, not 50 stocks. Correlation matrix is low rank.
2. **Brownian bridge:** Simulate components sequentially, condition on terminal values.
3. **Quasi-MC:** Sobol sequence handles high dimension better than random (if effective dimension is low).

Also use variance reduction: delta hedge as control variate (portfolio Delta known). And parallelize: 50 assets $\times$ 10k paths = 500k SDEs $\rightarrow$ GPU essential."

10. How would you parallelize MC? 16 cores available. **Answer:** "Embarrassingly parallel—each path independent. Two approaches:

1. **Path-level:** Each core simulates $N/16$ paths. Combine at end. Use multiprocessing.pool.
2. **Asset-level:** Each core handles subset of assets (if basket). Less common.

Implementation: Precompute random numbers (Sobol), broadcast to workers. Use shared memory for read-only parameters. **Pitfall:** Random seed: ensure streams are independent (use jump-ahead). Result: 10 $\times$ speedup on 16 cores (overhead). For 100 $\times$ speedup, use GPU (CUDA kernels)."

"FVA = Cost of Replacing Uncollateralized Exposure" **Story:** You're long a swap (positive MTM). If you hedge, you need to post collateral. You fund that collateral at 5% but receive OIS at 2%. **3% spread = cost.**

Key Points: - **No standard:** Each bank has its own funding curve - **Double counting:** If you include FVA, you must adjust CVA (risky discounting vs risk-free) - **Desk view:** Treasury charges desk for funding cost

FVA Controversy" FVA CVA. CVA = credit loss. FVA = funding spread. "FVA = EE × (funding spread) × maturity" I calculate FVA by discounting EE at funding curve vs OIS. The difference is the cost."

15. MVA & KVA: The New Kids

MVA (Margin Valuation Adjustment):

$$MVA = \int_0^T IM(t) \cdot (f_F - r_{coll}) \cdot dt$$

IM = Initial Margin (cost of posting). **Driver:** Uncleared margin rules.

"MVA is Initial Margin's Rent" **Memory:** IM is now required for uncleared trades. You post IM, can't rehypothecate it, and earn OIS. **Cost** = funding spread × IM amount.

KVA (Capital Valuation Adjustment):

$$KVA = \int_0^T RWA(t) \cdot CoC \cdot dt$$

CoC = Cost of capital (hurdle rate, 10-15%). **RWA** = Risk-weighted assets.

"KVA is Regulatory Capital Tax" **Story:** A trade consumes \$10M RWA. Capital requirement = 12.5% × RWA = \$1.25M. Cost = \$1.25M × 12% = \$150k/year. KVA = NPV of this cost.

XVA Hierarchy" CVA = credit, FVA = funding, KVA = capital, MVA = margin" Total XVA 0.5-5% of notional for long-dated" In interviews, I explain that XVA turns risk-neutral price into *cost of doing business*. Each desk pays its tax."

Part 5: Connecting Risk Metrics to Desks & Products

16. Risk Metric Per Product Type: The Desk View

Equities/FX Options: - **Primary:** Delta, Gamma, Vega, Vanna, Volga (Greeks) - **Risk:** Vega/VaR, smile shock (skew steepening) - **Stress:** Vol +20 points, spot gap -15%

"Equity Vol Desk: Short Vega = Short Vol of Vol" **Interview:** "Short vol book has negative Vanna (vol down when spot down). Stress: spot crash + vol spike = gamma loss + vega loss. Need to buy wings."

Rates Swaps/Swaptions: - **Primary:** DV01, PVBP, CS01 (credit spread), key-rate DV01 - **Risk:** IR VaR (rate shifts), basis risk (Libor-SOFR) - **Stress:** +200bp parallel shift, steepener

Credit Derivatives: - **Primary:** CS01, jump-to-default (JTD) - **Risk:** Spread VaR, correlation (CDO tranches) - **Stress:** +500bp CDS widening, correlation → 1

Structured Products: - **Primary:** Scenario-based PnL (no Greeks) - **Risk:** Model risk, gap risk, correlation risk - **Stress:** Custom scenarios (e.g., correlation breakdown for CDO)

17. How Traders See Risk Limits: The Escalation Game

Risk Limit Types:

- **Position limits:** Max notional (e.g., \$100M delta per name)
- **Greek limits:** Vega < \$500k per vol point